Java Operator Precedence

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When a calculation contains more than one operator, Java follows **order of operations** rules to decide which part to calculate first.

For example, multiplication happens before addition:

Example

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```
int result1 = 2 + 3 * 4;      // 2 + 12 = 14
int result2 = (2 + 3) * 4;    // 5 * 4 = 20

System.out.println(result1);
System.out.println(result2);
```

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Why Does This Happen?

In `2 + 3 * 4`, the multiplication is done first, so the answer is `14`.

Tip: Always use parentheses `( )` if you want to make sure the calculation is done in the order you expect. It also makes your code easier to read.

Order of Operations

Here are some common operators, from highest to lowest priority:

- `()` - Parentheses
- `*`, `/`, `%` - Multiplication, Division, Modulus
- `+`, `-` - Addition, Subtraction
- `>`, `<`, `>=`, `<=` - Comparison
- `==`, `!=` - Equality
- `&&` - Logical AND
- `||` - Logical OR
- `=` - Assignment

Another Example

Subtraction and addition are done from left to right, unless you add parentheses:

Example

```
int result1 = 10 - 2 + 5;     // (10 - 2) + 5 = 13
int result2 = 10 - (2 + 5);   // 10 - 7 = 3

System.out.println(result1);
System.out.println(result2);
```

Remember: Parentheses always come first. Use them to control the order of your calculations.

Exercise ?

Drag and drop the correct operator to add the numbers.

```
int sum = 5       3;
```

+ - * /

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